

Synoptic CB: Porewater Nutrients

September 2023 Samples

2025-12-22

Contents

0.1	Import Data & Clean	2
0.2	Assessing standard Curves	3
0.3	TEMPORARY CHUNK FOR SEPT 2023 - Run finished then restarted for dilutions	7
0.4	Dilution Corrections - ensure the latest dilution is kept	7
0.5	Performance Check	7
0.6	Analyze the Check Standards	9
0.7	Analyze Blanks	10
0.8	Analyze Duplicates	11
0.9	Spikes	12
0.10	Matrix Effects	13
0.11	Unit Converted Data Column Added (mg/L to uM)	13
0.12	Sample Flagging - Within range of standard curve	13
0.13	Pull out sample id information	13
0.14	Pulling Rhizon Samples	13
0.15	Check to see if samples run match metadata & merge info	14
0.16	Visualize Data	15
0.17	Export Processed Data	17

##Run Information

```
cat("Run Information: Input by User") #lets you know what section you're in
```

Run Information: Input by User

```
#set the run date & user name
run_date <- "01/25/2024"
sample_year <- 2023
sample_month <- 09
user <- "Stephanie Wilson"

#identify the files you want to read in
#read in as a list to accomodate ultiple runs in a month
NOx_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NOx_Sept2023_1.csv",
              "Raw Data/SEAL_COMPASS_Synoptic_NOx_Sept2023_2.csv",
              "Raw Data/SEAL_COMPASS_Synoptic_NOx_Sept2023_3.csv")
NH3_PO4_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_Sept2023_1.csv",
                  "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_Sept2023_2.csv",
                  "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_Sept2023_3.csv")

# Define the file path for QAQC log file - NO Need to change just check year
file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2023.csv"
final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_202309.csv"

#record any notes about the run or anything other info here:
run_notes <- "This run was stopped and restarted part way through due to
insufficient water and/or reaction segments.
There are fewer spikes for P04 and NH3 in this run due to stop and rerun.
NH3 peCheck is high, but other run metrics okay, so accepting this run."

#Set up file path for metadata
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2023.csv"

cat(run_notes)
```

```
## This run was stopped and restarted part way through due to
##   insufficient water and/or reaction segments.
##   There are fewer spikes for P04 and NH3 in this run due to stop and rerun.
##   NH3 peCheck is high, but other run metrics okay, so accepting this run.
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

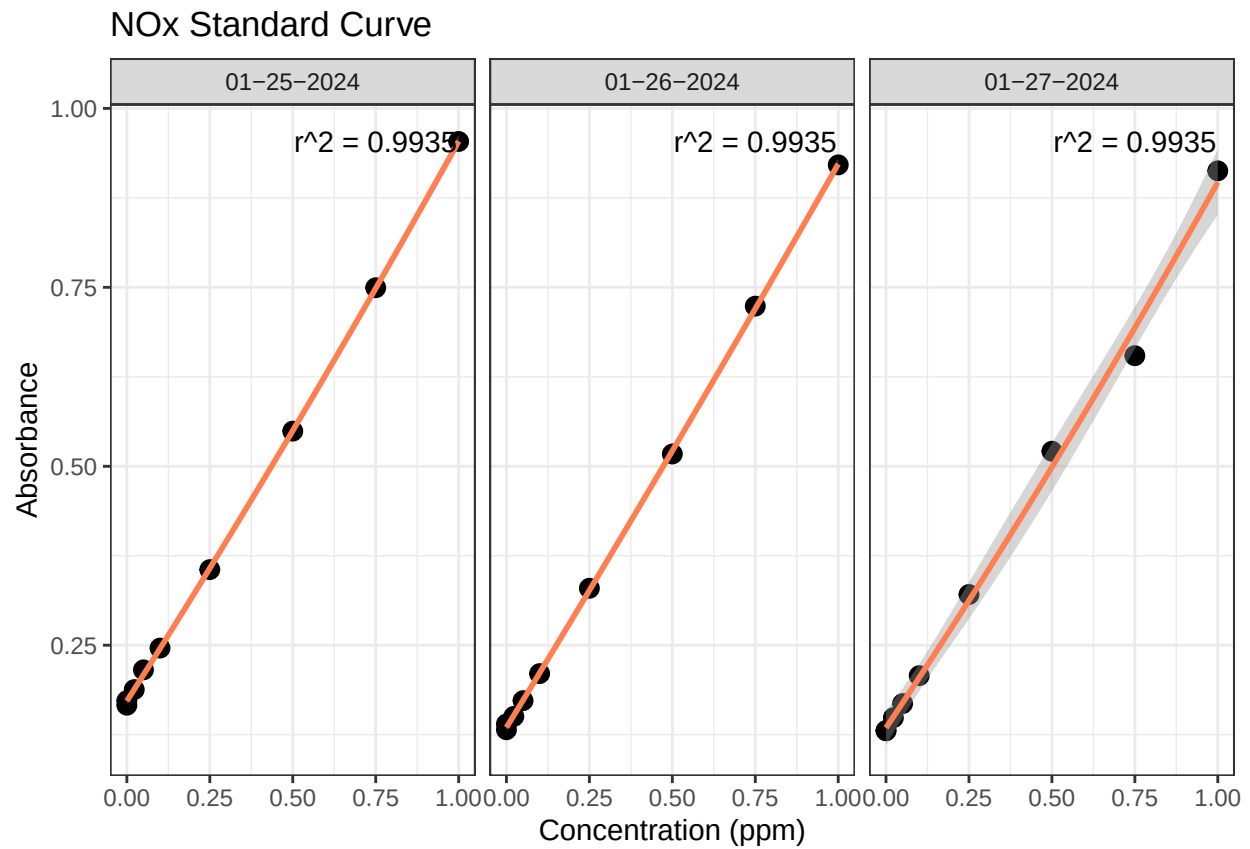
0.2 Assessing standard Curves

```
#Pull out standards data
```

```
## Assess Standard Curves
```

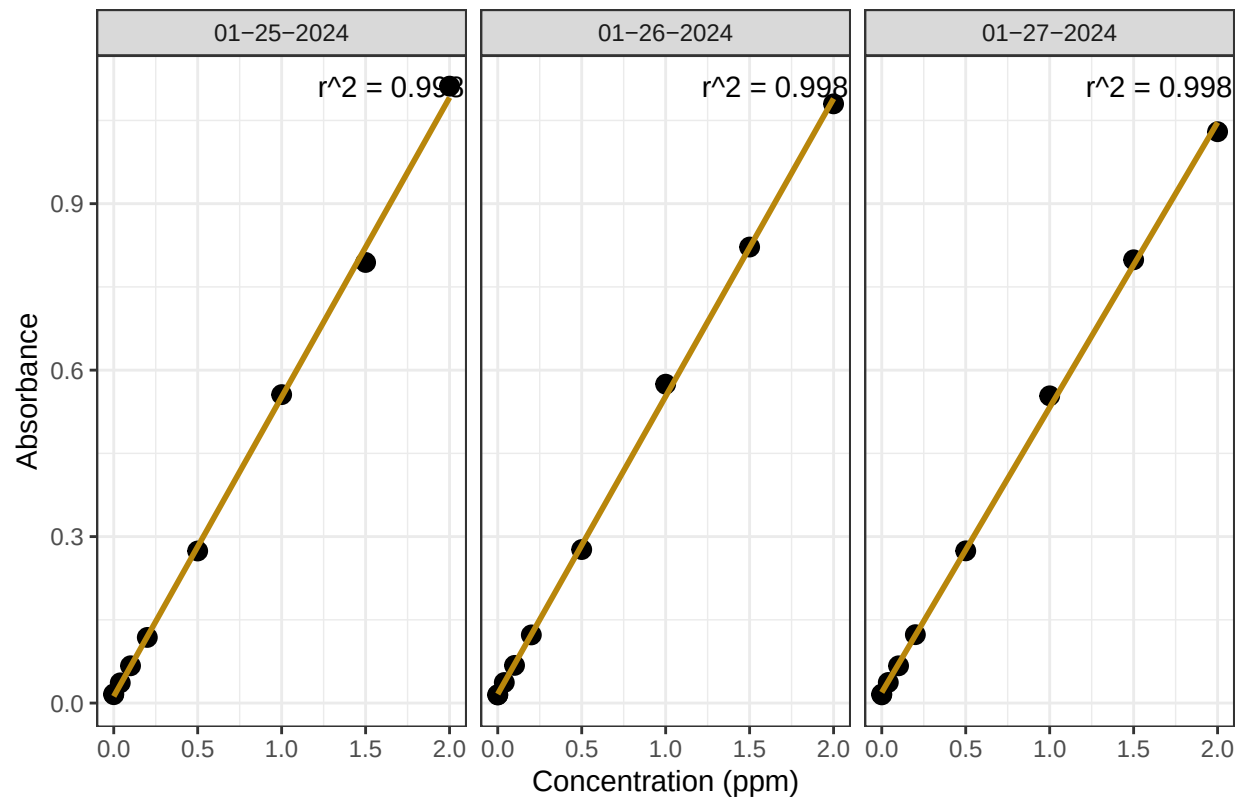
```
#Plot standards data
```

```
## Assess Standard Curves
```



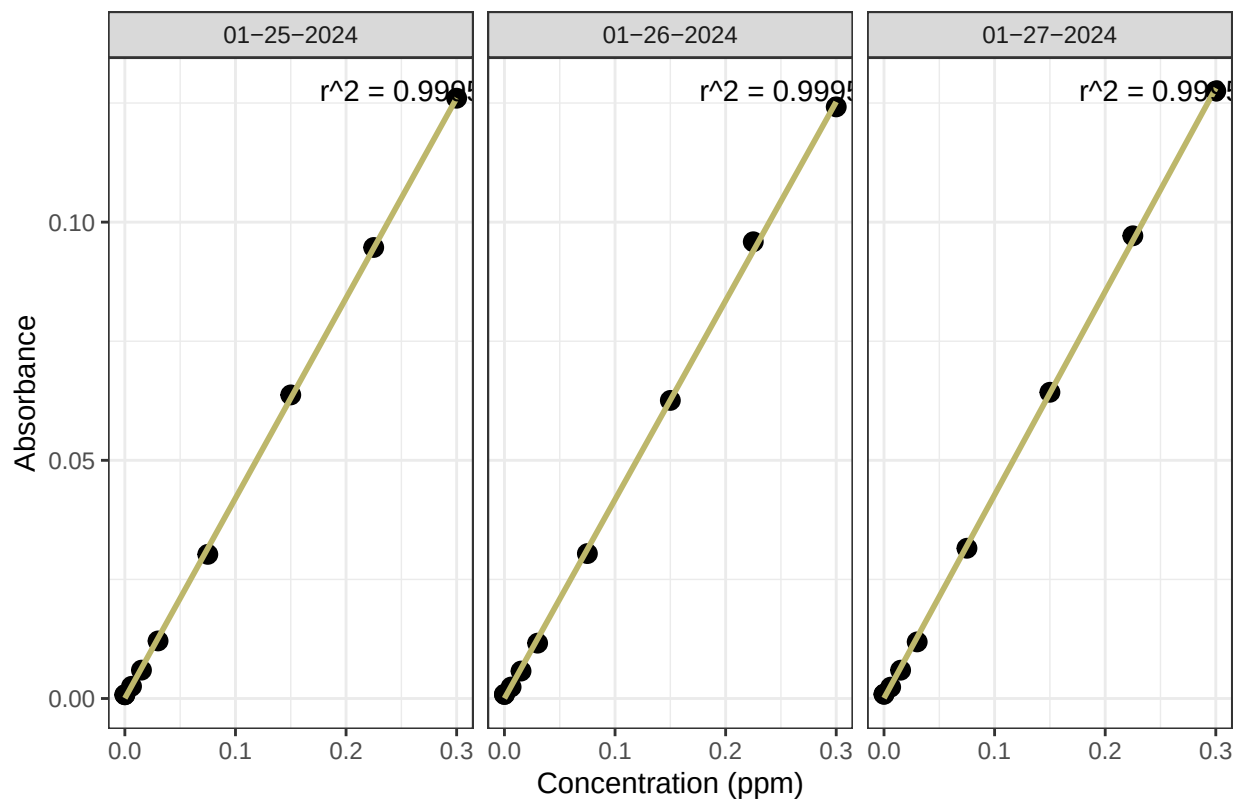
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



```
## 'geom_smooth()' using formula = 'y ~ x'
```

PO4 Standard Curve



```
## [1] "NOx Curve r2 GOOD - PROCEED"
```

```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "PO4 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

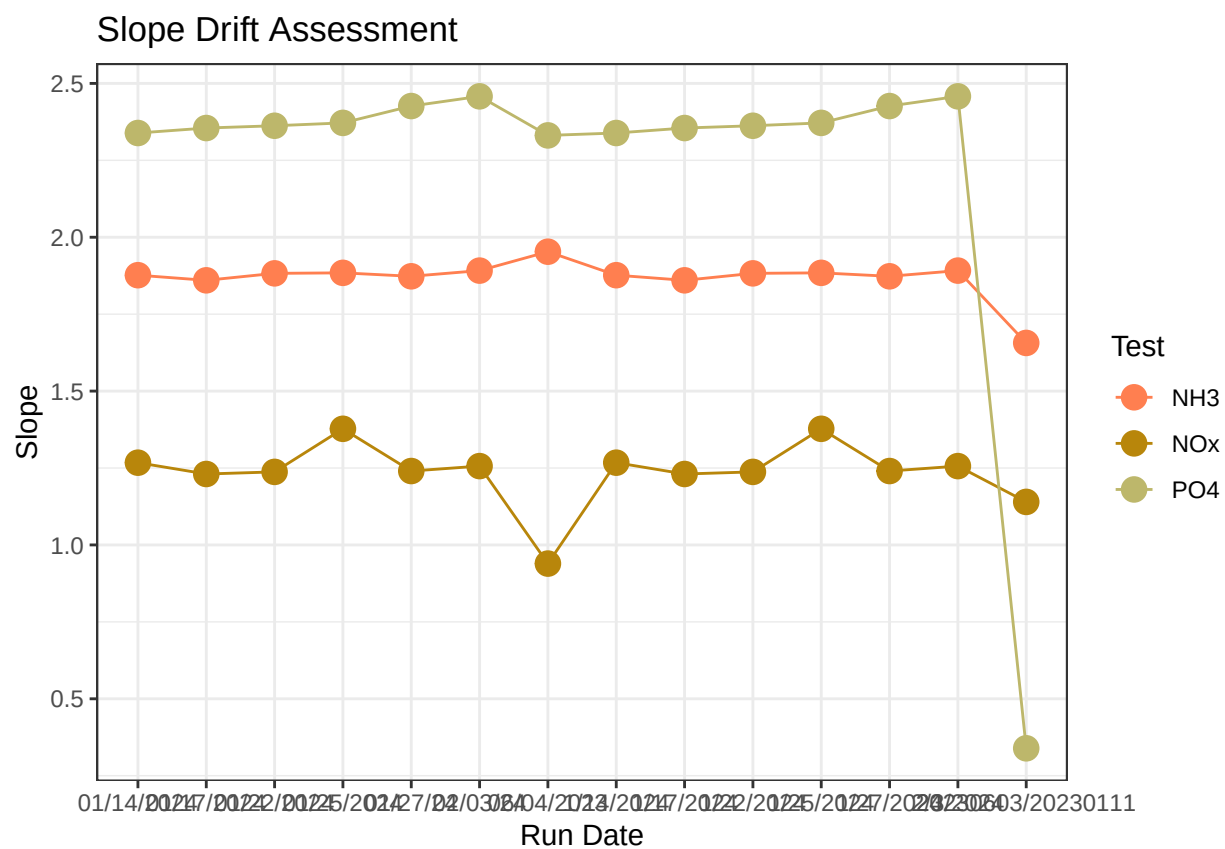


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.868
NOx	1.236
PO4	2.235

0.3 TEMPORARY CHUNK FOR SEPT 2023 - Run finished then restarted for dilutions

0.4 Dilution Corrections - ensure the latest dilution is kept

Dilution Corrections

No duplicated samples.

0.5 Performance Check

[1] "NOx pe Check has a % Difference <25% - PROCEED"

Run mean = 0.738716

Expected = 0.706

[1] "NH3 pe Check has a % Difference >25% - REASSESS"

Run mean = 1.531811

Expected = 0.948

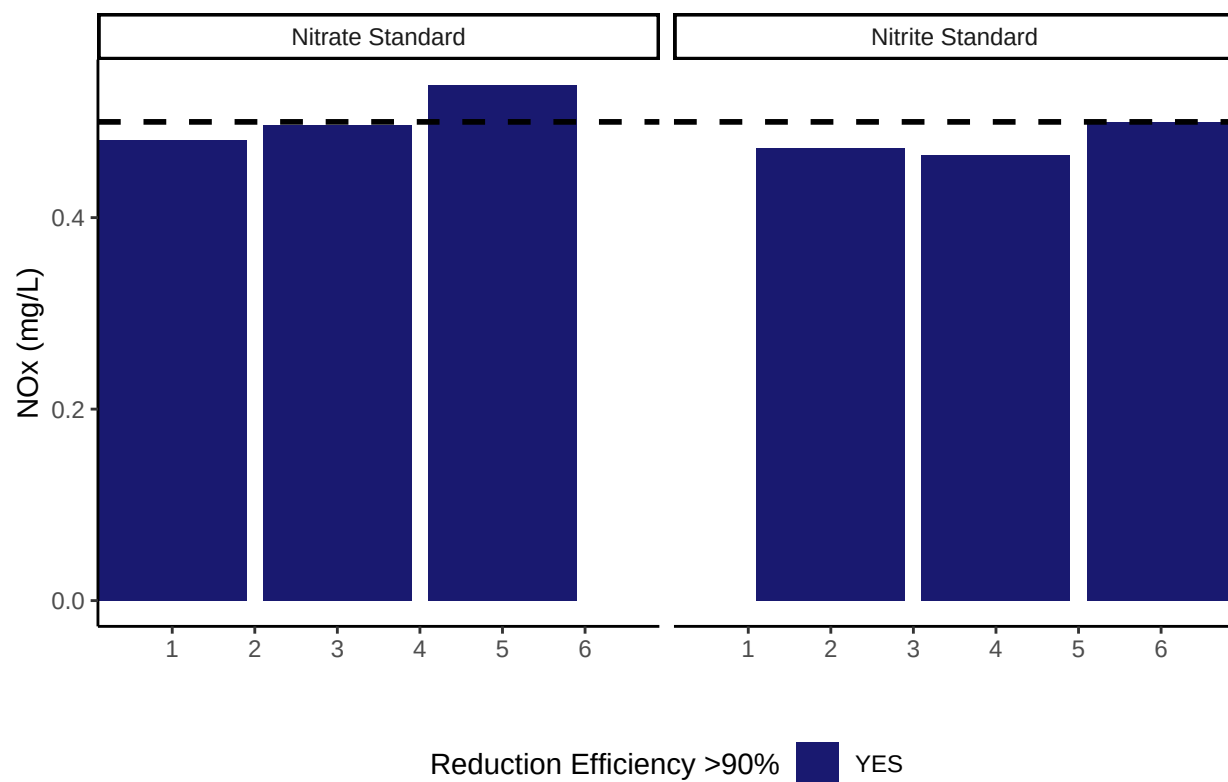
[1] "P04 pe Check has a % Difference <25% - PROCEED"

Run mean = 0.864723

Expected = 0.818

#Check NOx Reduction Efficiency

Assess Reduction Efficiency



```
## [1] "Mean NOx Reduction Efficiency >95% - PROCEED"
```

```
## [1] 98.47613
```

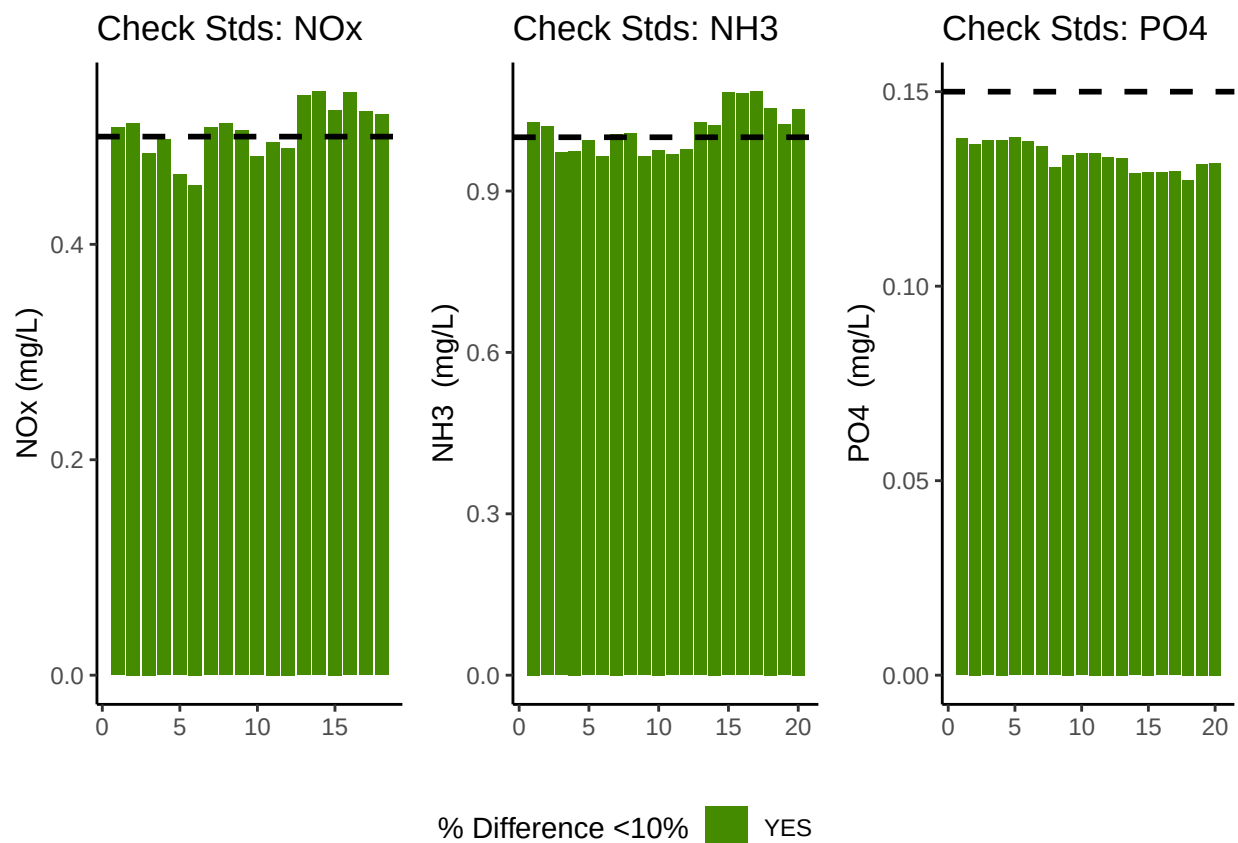

0.6 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NOx Check Standard RSD within Range - PROCEED"
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```

```
## [1] "PO4 Check Standard RSD within Range - PROCEED"
```



```
## [1] ">60% of NOx Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of PO4 Check Standards are within range of expected concentration - PROCEED"
```

0.7 Analyze Blanks

```
## Assess Blanks
```

```
## [1] ">60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 detection limit - PROCEED"
```

```
## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"
```

```
## [1] ">60% of PO4 Blanks are below the lower 25% quartile of samples- PROCEED"
```

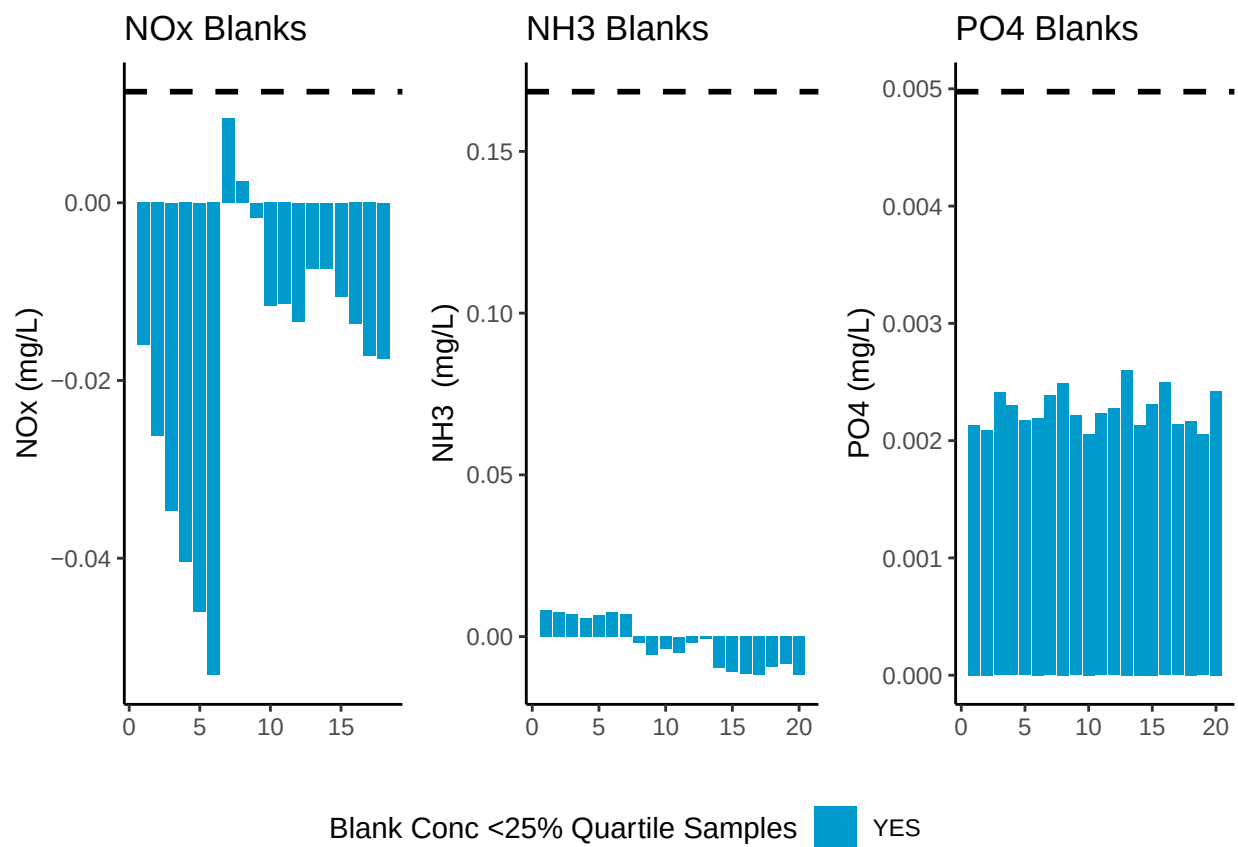


Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	-0.0176
NH3	-0.0022
PO4	0.0023

0.8 Analyze Duplicates

```
## Analyze Duplicates
```

```
## [1] ">60% of NOx Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">60% of PO4 Duplicates have a CV <10% - PROCEED"
```

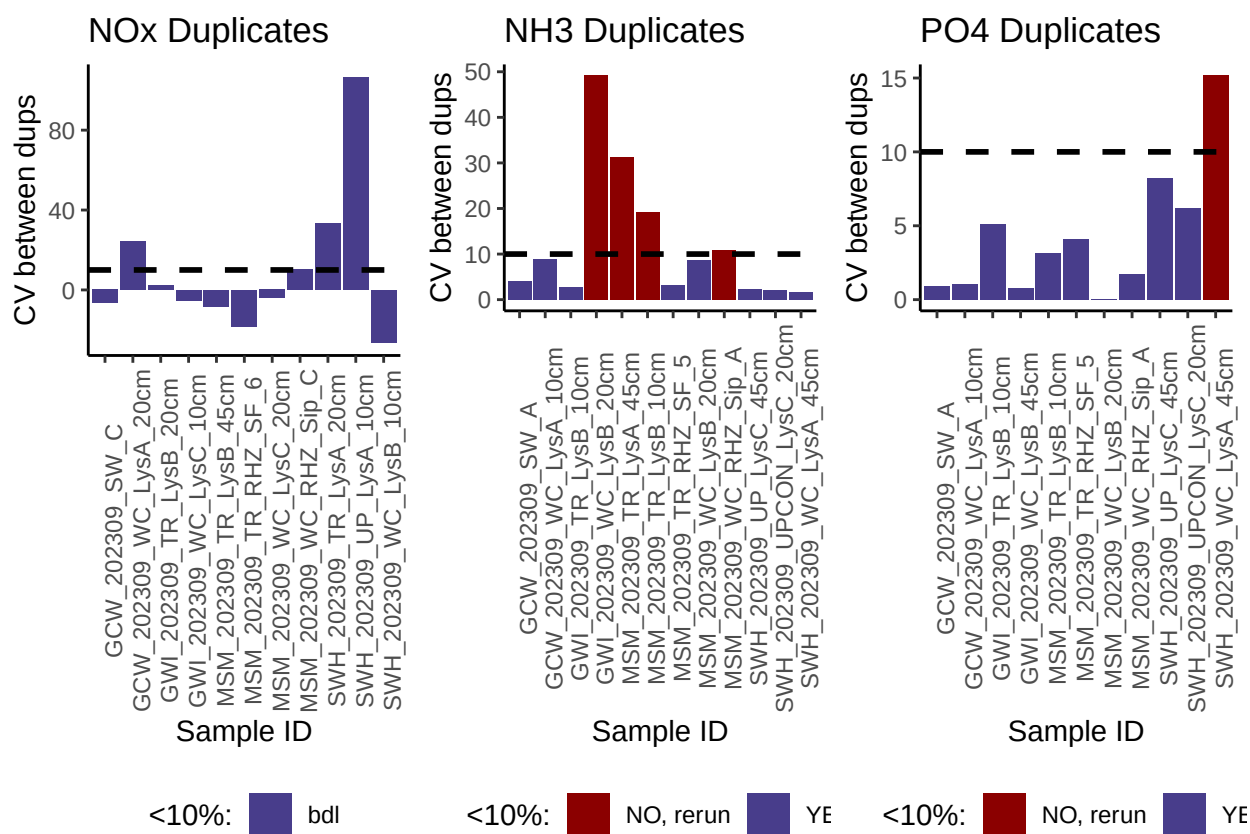
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use 'linewidth' instead.
```

```
## This warning is displayed once every 8 hours.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```

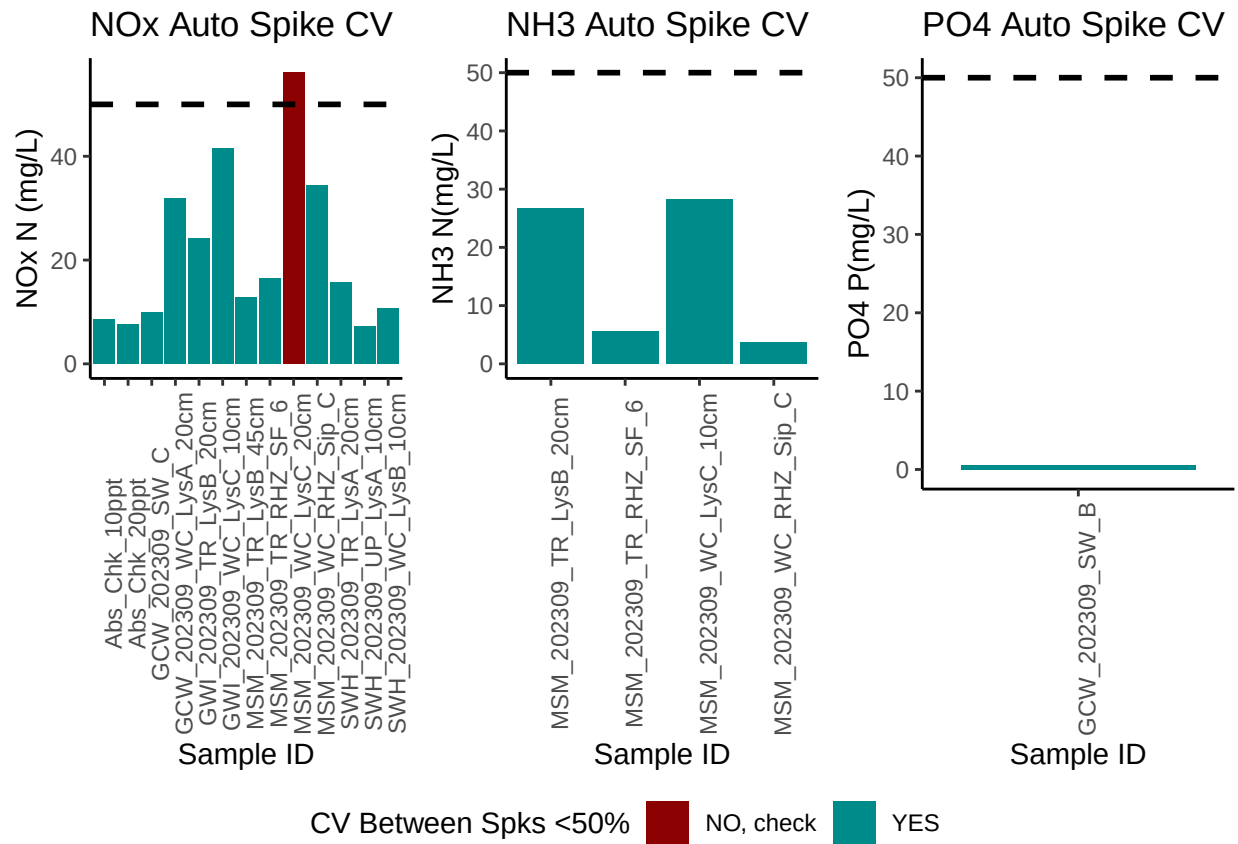


0.9 Spikes

[1] ">60% of Spikes have a CV <50% - PROCEED"

[1] ">60% of Spikes have a CV <50% - PROCEED"

[1] ">60% of Spikes have a CV <50% - PROCEED"



0.10 Matrix Effects

```
## [1] "NO NOx Matrix Effect, PROCEED"
```

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

```
## [1] "NO PO4 Matrix Effect, PROCEED"
```

0.11 Unit Converted Data Column Added (mg/L to uM)

0.12 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

0.13 Pull out sample id information

```
## Sample Processing
```

```
## Warning: There was 1 warning in 'mutate()'.  
## i In argument: 'Run_Time = lubridate::mdy_hm(Run_Time)'.  
## Caused by warning:  
## ! All formats failed to parse. No formats found.
```

```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 36 rows [18, 19, 20, 42,  
## 43, 44, 77, 78, 79, 101, 102, 103, 121, 122, 123, 138, 139, 140, 157, 158,  
## ...].
```

0.14 Pulling Rhizon Samples

```
# Filter rhizon and peeper samples  
df_rhizon <- df_all %>%  
  filter(str_detect(Sample_Name, "RHZ"))  
df_peep <- df_all %>%  
  filter(str_detect(Sample_Name, "PPR"))  
  
# Timestamp for backups  
timestamp <- format(Sys.time(), "%Y-%m-%d_%H%M")  
  
# Paths  
folder_path <- file.path("Raw Data", "Rhizon+Peeper")  
dir.create(folder_path, recursive = TRUE, showWarnings = FALSE)  
  
rhizon_main <- file.path(folder_path, "rhizon_data.csv")  
peeper_main <- file.path(folder_path, "peeper_data.csv")  
  
rhizon_backup <- file.path(folder_path, paste0("rhizon_data_", timestamp, ".csv"))  
peeper_backup <- file.path(folder_path, paste0("peeper_data_", timestamp, ".csv"))  
  
# Write timestamped backups
```

```
write.csv(df_rhizon, rhizon_backup, row.names = FALSE)
write.csv(df_peep, peeper_backup, row.names = FALSE)

# Overwrite the main files with latest data
write.csv(df_rhizon, rhizon_main, row.names = FALSE)
write.csv(df_peep, peeper_main, row.names = FALSE)

## ^^ I think there is a cleaner way to write this out, but this should work for now ^^
```

0.15 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

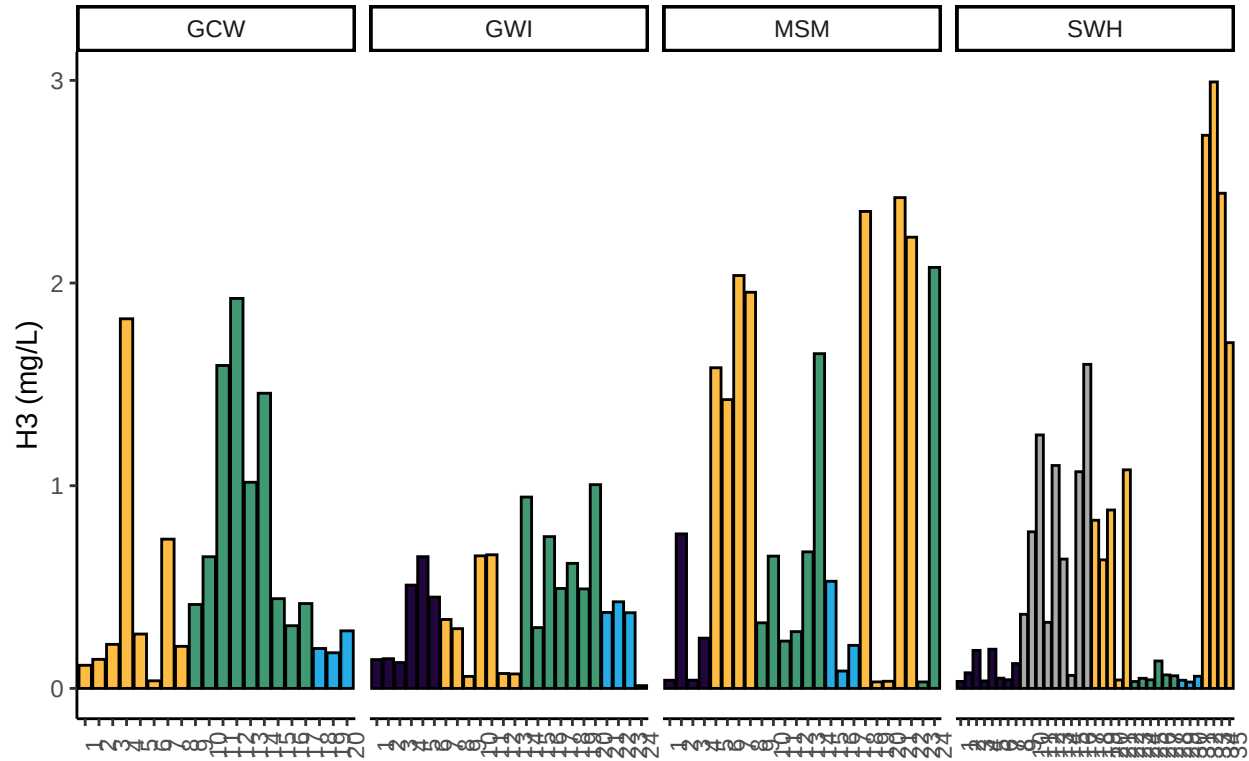
```
## All sample IDs are present in metadata.
```

```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 36 rows [18, 19, 20, 42,
## 43, 44, 77, 78, 79, 101, 102, 103, 121, 122, 123, 138, 139, 140, 157, 158,
## ...].
```

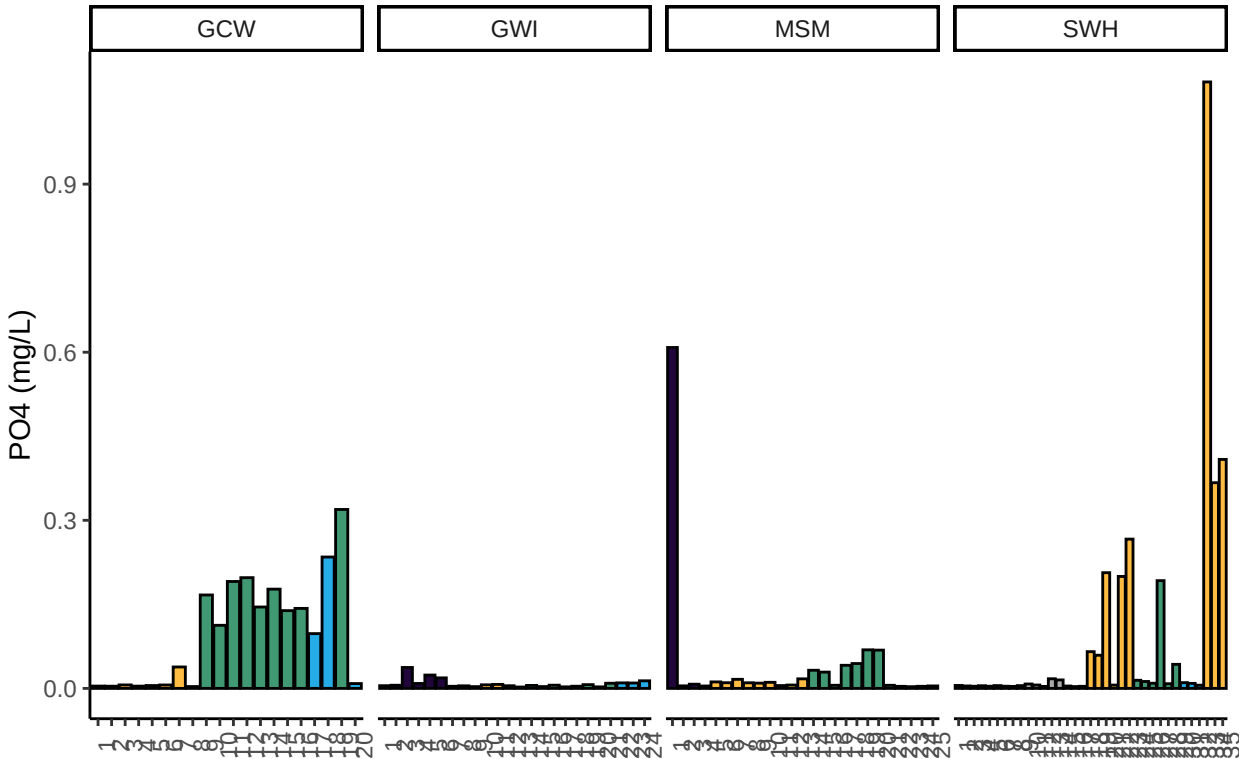
Visualize Data



Porewater NH3



Porewater PO₄



0.17 Export Processed Data

#end